

How Much of the Kalamazoo Promise Was Capitalized in the Local Housing Market?

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Abstract

This paper examines whether the Kalamazoo Promise, a full-ride college scholarship, helped attract people to Kalamazoo. Using ACS data from 2005–2024 and a difference-in-differences framework, we compare Kalamazoo to similar Michigan counties. Results show a statistically significant 5.4% increase in mortgage payments, indicating capitalization into home values, alongside a 2.6% decline in rents. These findings suggest the policy increased demand for owner-occupied housing while affecting rental markets differently, highlighting how place-based educational benefits are unevenly reflected across housing segments.

Introduction

Residential real estate markets play a central role for people and communities alike. For individuals, homeownership can be a source of wealth accumulation and household financial stability. At the community level, it matters for local fiscal health through property tax revenues. Which funds public services, including schools, infrastructure, and safety. In the United States, real estate is one of the largest asset classes for American households, making the housing market the central determinant of deciding the economic health of a community and a nation.

Governments can implement new place-based policy, meaning a policy tied to a specific location. It can alter the desirability of that location by improving the economic conditions of a specific area. Examples include enterprise zones, urban renewal programs, and education initiatives. Because the benefits are location specific, they influence where households choose to live. Families with school-age children are drawn to areas that have stronger school programs. This increased demand puts upward pressure on local housing prices and rents, a process economists refer to as amenity capitalization. Academic research has explored this topic. One of the earliest and most studied examples of such a program is the Kalamazoo Promise. LeGower & Walsh (2017) and Bartik & Sotheland (2015) both find evidence that a Promise-style program affects housing prices and migration patterns.

Kalamazoo in the early 2000s was a post-industrial city experiencing population decline and falling school enrollment, like many other small Michigan cities. Then, in 2005, the Kalamazoo Promise was launched by a group of anonymous private donors. It offers Kalamazoo Public Schools (KPS) graduates who start in KPS in kindergarten and graduate high school in a KPS up to 100% tuition coverage at Michigan public colleges and

universities. There is a sliding scale determining the scholarship amount based on when a student started in the KPS system. Table 1 describes the tuition covered based on when a student started in the KPS system, with the last year of eligibility being 9th grade. Critically, because scholarship eligibility is tied to continuous enrollment within the KPS district, the program creates a direct residential incentive. Families seeking to maximize their scholarship benefit have a reason to move into or remain within the district boundaries (Bartik & Sotherland 2015). A dynamic that may translate into increased housing demand and upward pressure on local prices.

Existing studies have examined the Promise's effects on school enrollment, migration, and housing prices broadly. LeGower and Walsh (2017) find housing price effects concentrated in higher-quality school neighborhoods. Bartik and Sotherland (2015) find reduced out-migration and modest population growth. Miller (2018) finds little effect on housing prices. While this body of work establishes a strong foundation, there are two limitations that warrant further research. First, prior studies have focused on property values and used controls that might not adequately satisfy the parallel trends assumption. Second, most existing analyses use data from the early post-treatment period. This paper addresses both of those concerns. By aiming to answer whether the Kalamazoo Promise caused a measurable change in median mortgage payments and rental prices within Kalamazoo relative to comparable areas, and extending the analysis through 2024.

This paper uses a Difference in Differences (DID) framework comparing Kalamazoo to a set of control areas. Over time by comparing how Kalamazoo's housing costs changed relative to control areas before and after 2005 isolate the causal effect of the Promise. The research can isolate the causal effect of the Promise from broader trends affecting all areas equally. The data comes from American Community Survey (ACS) and covers mortgage payments and rental prices, and it runs through 2024. The controls were determined by Michigan counties such as Ingham, Berrien, Calhoun, Eaton and Clinton, and Jackson that share socioeconomic, and demographic backgrounds to Kalamazoo to ensure the parallel trends assumption holds (ACS 2005, 2024).

The results suggest that the Kalamazoo Promise was associated with an increase in median monthly mortgage payments and an increase in rental prices relative to control areas, effects that are statistically significant at the 0.1% level. The remainder of the paper proceeds as follows. Section II reviews previous literature of the Promise. Section III outlines the empirical data and methodology. Section IV presents the results, and Section V concludes.

Literature Review

LeGower and Walsh (2017) provide the first large scale evaluation of Promise programs by analyzing a broad cross section of 21 programs nationwide to determine their impact on K-12 enrollment and 8 programs to measure housing price effects. They use a DID approach comparing affected Promise zones to their immediate geographic surroundings. Specifically, examining how different programs feature merit requirements influence outcomes. They find an average enrollment increase of 4% across all programs (LeGower & Walsh, 2017). Programs that allowed flexible college choices like Kalamazoo saw the largest gains at 8% (LeGower & Walsh, 2017). For housing, they find a 7% to 12% increase in residential property values within the first three years of a program announcement (LeGower & Walsh, 2017). Also, price increases were concentrated in the upper half of the housing market, suggesting the benefit was mostly received by middle- and upper-income households. The present paper builds on this work by focusing specifically on Kalamazoo, extending analysis through 2024, and examining median monthly mortgage payments and rental prices as distinct outcomes beyond aggregate property values.

Bartik & Sotheland (2015) measure the short term impact of eight different Promise programs including Kalamazoo, Pittsburgh, and New Haven on local migration patterns and residential property values. They utilized panel data from the ACS (2005- 2013), analyzing both broad commuting zones and smaller migration Public Use Microdata Areas (PUMAs). They found that Promise programs significantly reduce out migration. This effect was greatest for households with children. They report a temporary spike in families moving into the district during the year of the announcement, though this trend was not sustained in later years. While they also observe no strong evidence that the Promise increased housing prices even though prices rose. They explain that they were already trending upward prior to the scholarship, making it difficult to credit the scholarship alone. For Kalamazoo it is estimated that reduced out migration led to a 1.7% increase in population, and approximately 1,900 permanent jobs in a district like Kalamazoo (Bartik & Sotheland, 2015). They conclude that Promise zones are more effective at retaining existing families rather than a Promise luring in new families. The present paper extends this work by using a DID approach that explicitly compares Kalamazoo's pre treatment to their post treatment in accordance with other counties, providing a stronger basis for isolating the causal effect of the Promise on housing costs.

Miller (2018) examines the Kalamazoo Promise primarily as an assessment of the scholarship as a lure for families and a tool for economic development. To determine this, Miller analyzed enrollment data from the Michigan Center for Educational Performance

and Information (CEPI) for the years 2002 through 2009. Miller then used DID to compare enrollment changes in KPS against three separate comparison groups within Kalamazoo County. Miller's key findings include robust evidence regarding enrollment gains. KPS schools increased by about 1,000 students in the first year with gains spread across both high-poverty and low-achieving schools (Miller 2018). To assess the capitalization hypothesis, the idea that educational amenities increase property values, Miller used the same method using residential property sales in Kalamazoo County from 2001 to 2010. Miller's property results although proved inconclusive, where Miller found no evidence of an increase and even suggested a 7.8% to 7.9% decrease (Miller 2018). A potential limitation of this analysis is that Miller's data coincides with the 2008 financial crisis that impacted the more vulnerable Kalamazoo home than the surrounding areas of Kalamazoo. The present paper addresses this concern by extending the analysis through 2024 and selecting control areas that more closely mirror Kalamazoo's economic conditions.

Beasley and Alam (2025) provide an updated assessment specifically targeting the impact of residential property values. They wanted to simulate how families weigh the tradeoff between a future scholarship and current school quality. Their research suggests that a meaningful increase in home prices in a distressed area must be sufficiently large to offset any lower neighborhood quality. To find a similar comparison to KPS homes, they implemented a Propensity Score Matching (PSM) to better evaluate KPS homes with other homes in Michigan. Unlike Miller (2018) who compared Kalamazoo to its surrounding neighborhoods, they tried to match KPS homes to homes in Michigan that were structurally and educationally similar to eliminate bias. They find a significant positive capitalization effect of 12.4% to 15%, which equates to an average home value gain of roughly \$15,207 (Beasley & Alam 2025). The present paper builds on this work by applying a DID framework similar to theirs, extending the analysis through 2024, and focusing specifically on median monthly mortgage payments and rental prices as distinct housing cost outcomes.

Data Description

We use data from the ACS, which provides individual and household information on housing costs, demographics, and socioeconomic characteristics. Our dataset (Michigan) has variables on mortgage payments, rent, income, and housing characteristics.

The sample covers the period from 2005 to 2024, allowing us to analyze the housing outcomes before and after the introduction of the Kalamazoo Promise in late 2005 (with post-treatment beginning in 2007).

The unit of observation is the household (or housing unit), as housing costs such as rent and mortgage payments are measured at that level.

We restrict the sample to occupied housing units by keeping observations with mobility status, which excludes vacant units. We also drop observations with missing values for both rent and mortgage payments to ensure accurate measurements of housing costs.

The dependent variables are monthly housing costs. We measure these using monthly mortgage payments (*mr gp*) and monthly rent payments (*gr ntp*). The treatment (*treat*) is estimated using a DID framework.

$$gen\ kzoo = (puma = \text{Kalamazoo})$$

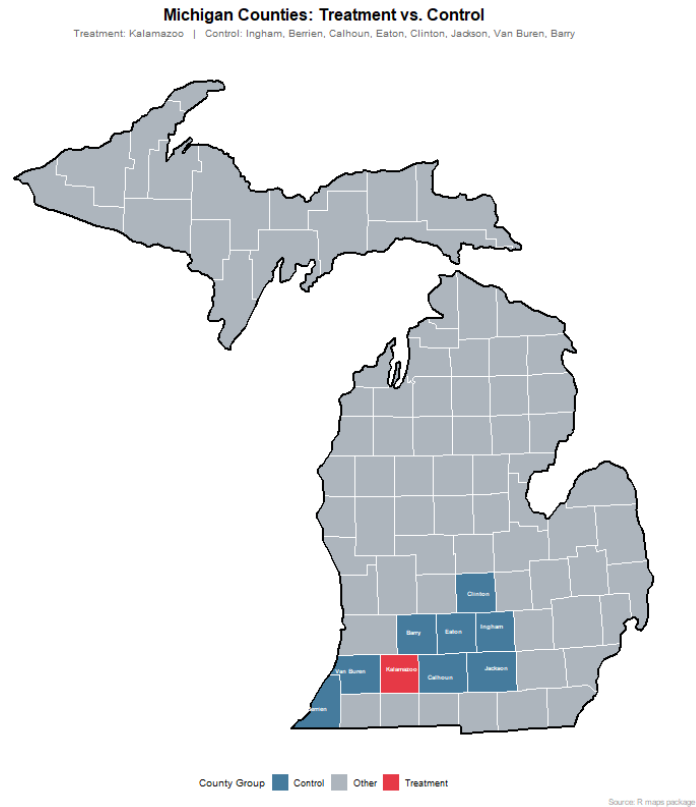
$$gen\ post = (year \geq 2007)$$

$$gen\ treat = kzoo \times post$$

The Kalamazoo variable (*kzoo*) equals 1 for households in Kalamazoo and 0 otherwise, capturing the treated group. The post-treatment variable (*post*) equals 1 for years 2007 and later, showing the period after the Kalamazoo Promise. The interaction term (*treat*) combines the two variables (*kzoo* $\times$ *post*) and explains households in Kalamazoo after the promise.

We control household and housing characteristics, including household income (*hincp*), age (*agep*), household size (*np*), number of bedrooms and rooms (*bdsp rm sp*), and demographic controls such as race (*rac1p*), gender (*sex*), education (*schl*), and marital status (*mar*).

Map 1



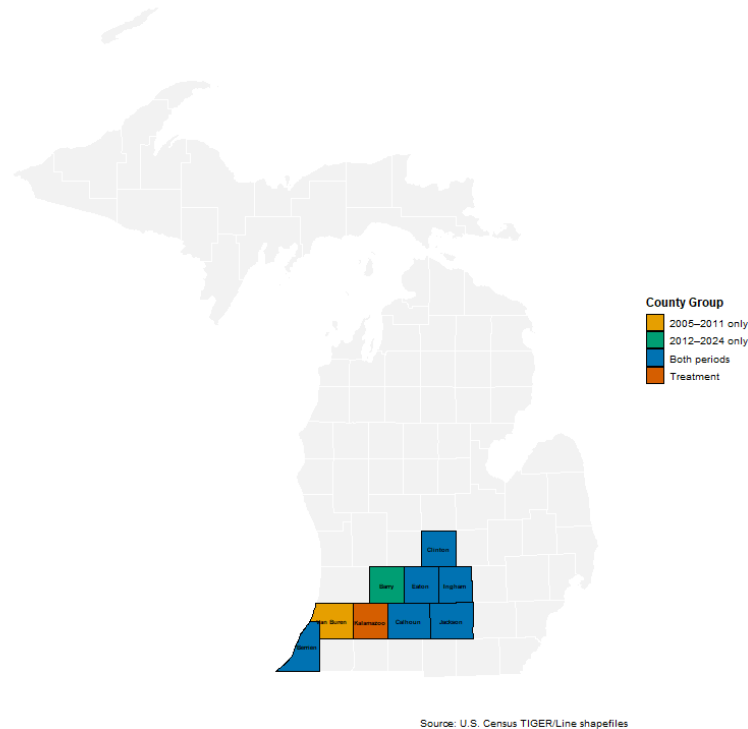
Source: R Map Package

One of the main difficulties we encounter with our dataset (Michigan) is the use of PUMA, which makes it harder to identify cities or counties precisely. To choose our control group, we follow Bartik et al. (2014) by selecting counties that are close to Kalamazoo and have similar labor market conditions, demographics, and housing characteristics. Our control group includes Barry, Berrien, Calhoun, Clinton, Eaton, Ingham, and Jackson counties, which serve as a suitable comparison to our treatment.

Map 2

Michigan Counties by Treatment and Control Periods

Treatment: Kalamazoo County
Comparison counties vary by period



Source: R Map Package

The differences observed in the map arise from changes in PUMA and their correspondence to counties over time. Because PUMA are geographic units designed for statistical purposes, they do not align perfectly with county boundaries, and some counties are grouped differently across periods. These changes reflect redistricting associated with the decennial census, which necessitates periodic realignment of geographic boundaries to maintain consistent population thresholds and improve the accuracy of survey estimates. For the 2005–2011 ACS data, Van Buren County was included in PUMA 2602, which also encompassed outer Kalamazoo County. Beginning in 2012, this designation changed to PUMA 2101, which included only outer Kalamazoo County. A similar shift occurred with Calhoun County: from 2005–2011, it was part of PUMA 2800, but after 2012 it was reassigned to PUMA 2000, which then included Calhoun County.

Summary Statistics

Table 2 shows sum statistics for housing costs and household characteristics across treatment and control before and after the policy. Prior to the policy, mortgage payments are slightly lower in Kalamazoo (\$844) compared to the control group (\$860),

and a similar pattern holds for rent, with Kalamazoo (\$701) below the control group (\$732). After the policy, housing costs increase in both groups, but the increase is higher in Kalamazoo. Mortgage payments rise to \$926 in Kalamazoo compared to \$874 in the control group, while rent increases to \$903 in Kalamazoo and \$897 in the control group.

Household income also increases over time in both groups, with Kalamazoo experiencing a slightly larger rise, which may contribute to higher housing demand. Other characteristics such as age, household size, and housing structure (bedrooms and rooms) remain similar across groups and over time, suggesting that the composition of households does not change much.

Overall, these patterns suggest that while housing costs increase over time in both areas, Kalamazoo experiences somewhat larger increases after the policy. However, these differences should be interpreted with caution, as they do not count for the underlying trends or other factors, motivating the use of a DID approach.

Figure 1

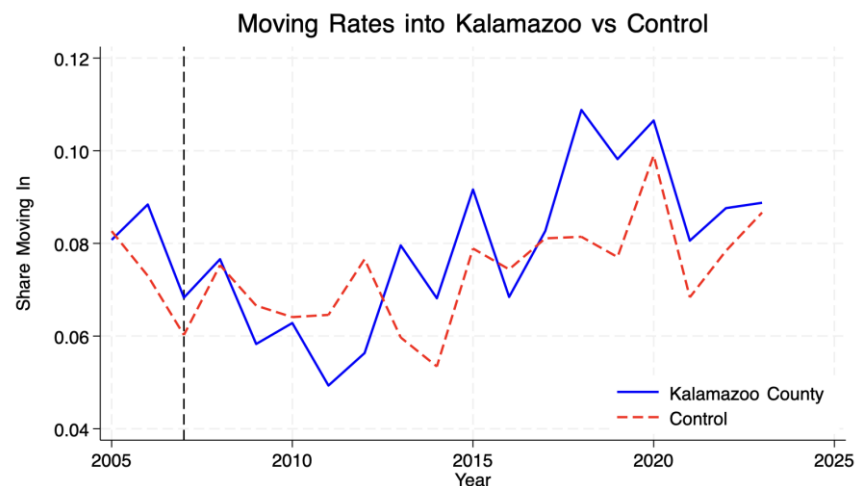


Figure 1 shows the share of households moving into Kalamazoo compared to the control group over time. Before the policy, moving rates in Kalamazoo and the control group follow similar patterns, suggesting comparable conditions prior to treatment. After 2007, moving rates in Kalamazoo increase slightly and become more volatile, sometimes exceeding those in the control group. However, these differences remain small. Overall, the figure suggests that there is some increase in migration, but it is modest and unlikely to be the main driver in housing outcomes.

Empirical Methodology

We use a DID approach to estimate the effect of Kalamazoo Promise on housing costs. This approach compares changes over time before and after the policy between treated (Kalamazoo) and control group, allowing us to isolate the policy's effect while controlling time trends and baseline differences across areas.

Regression Equations

$$\begin{aligned}\ln(Mortgage_{it}) = & 4.502 + 0.054 Treat_{it} + 0.144 \ln(Income_{it}) \\ & + 0.004 Age_{it} - 0.00004 Age_{it}^2 \\ & - 0.000 HouseholdSize_{it} + 0.097 Bedrooms_{it} + 0.040 Rooms_{it} + FE + \varepsilon_{it} \\ \ln(Rent)_{it} = & 6.180 - 0.026 Treat_{it} + 0.033 \ln(Income)_{it} \\ & - 0.014 Age_{it} - 0.00014 Age_{it}^2 \\ & + 0.038 HouseholdSize_{it} + 0.099 Bedrooms_{it} + 0.022 Rooms_{it} + FE + \varepsilon_{it}\end{aligned}$$

The coefficient on the treatment variable ($Treat$) measures the casual effect of the Kalamazoo Promise on housing costs. Since the dependent variable is in logarithms, β_1 can be interpreted approximately as a percentage change.

The coefficients from these estimating equations allow us to evaluate the impact of Kalamazoo Promise on housing outcomes. In particular, the coefficient on the treatment variable ($Treat$) captures the causal effect of being in Kalamazoo County after the introduction of the program, relative to the control group. Because the dependent variables are in logarithmic form, these coefficients can be interpreted as percentage changes. For example, a positive and statistically significant coefficient in the mortgage regression (0.054) indicates that the policy increased mortgage payments, while a negative coefficient in the rent regression (-0.026) suggests a decrease in rental costs. The control variables (such as income, age, and housing characteristics) help isolate this effect by accounting for differences in household and housing conditions, ensuring that the estimated treatment effect reflects the impact of the policy rather than other factors.

Potential Concerns

A key concern is migration into Kalamazoo following the policy, which may increase housing demand and bias estimates. Selection bias may arise if households moving into

Kalamazoo differ systematically from those in the control group. However, using mobility (mv) as a proxy, we find only modest changes after the policy, suggesting that large compositional shifts are unlikely to drive the results. Additionally, measurement of errors in self-reported housing costs may affect results. To mitigate this risk, we excluded observations with reported mortgage payments less than \$50 and rent below \$10.

Results

Figure 2

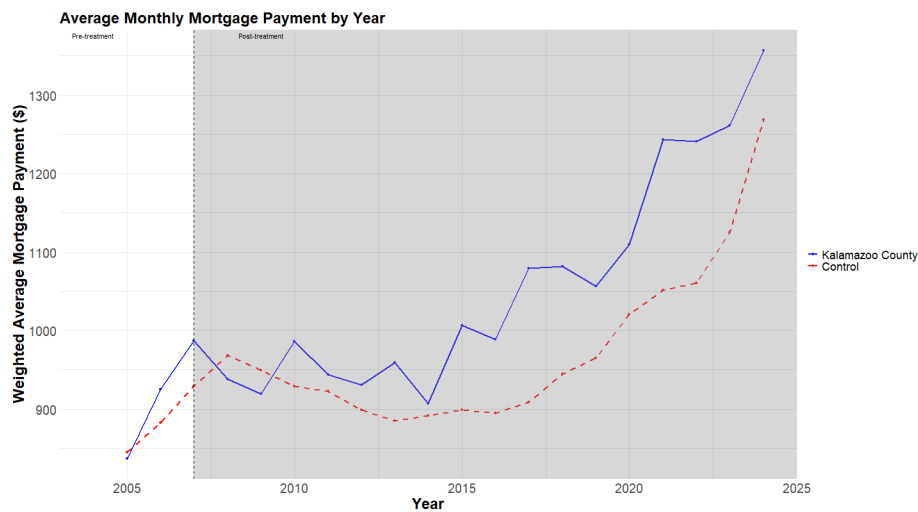


Figure 3

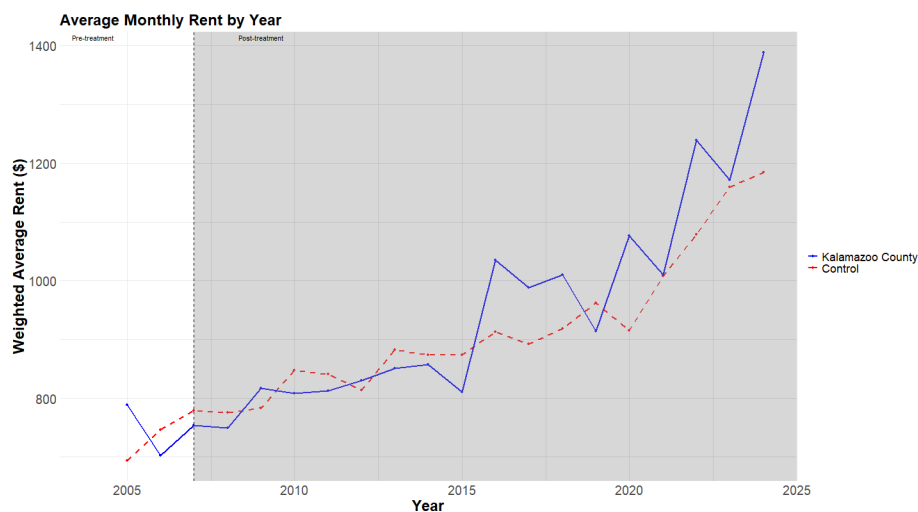


Figure 2 and Figure 3 present trends in average annualized monthly gross rent and mortgage payments for Kalamazoo County and a control variable representing

other counties exhibiting similar characteristics to Kalamazoo. However, although the Kalamazoo Promise was introduced in 2005, the treatment period is defined as beginning in 2007 to account for adjustment lags, as residential mobility and housing market responses are not instantaneous and may take time to be reflected in observed outcomes. Prior to 2007, both groups exhibit relatively similar trajectories, supporting the parallel trends assumption required for a DID framework. The absence of major divergence in the pre-treatment period strengthens the credibility of attributing post 2007 differences to the policy rather than pre-existing trends.

Following the onset of the treatment period, the mortgage payment graph shows a clear and sustained divergence between Kalamazoo County and the control group. Mortgage payments in Kalamazoo begin to rise more rapidly, with the gap widening notably in the later years of the sample. This pattern is consistent with a positive treatment effect, suggesting that the Kalamazoo Promise increased demand for owner-occupied housing, which in turn drove up home prices and associated mortgage costs. The persistence and magnitude of this divergence indicate that the policy is likely capitalized into housing values over time.

In contrast, the rent graph displays a more modest and less consistent divergence between Kalamazoo and the control group after 2007. Although rents in Kalamazoo occasionally exceed those in the control group, the differences are smaller and not as persistent as those observed for mortgage payments. This suggests that the treatment effect on rental markets is weaker, potentially reflecting differences in market dynamics, such as more elastic supply in rental housing or a stronger response among prospective homeowners than renters. Overall, the results imply that the Kalamazoo Promise had a more pronounced effect on housing prices than on rental costs.

Our regression analysis table is in the appendix section under Table 3, and we will use the values from our DiD models throughout the following section to answer our question whether the Kalamazoo Promise affected housing pricing. The treatment coefficient, Kalamazoo County and the Kalamazoo promise, is statistically significant for both regressions at the 0.1% level. In our mortgage regression, the treatment coefficient is positive (0.054), indicating that the Kalamazoo Promise increased monthly mortgage payments in the treatment group. In contrast, our rental regression shows a negative treatment coefficient (-0.026), indicating a decrease in monthly rent reduction following implementation. The high level of statistical significance, $p < 0.001$, provides strong evidence that these effects are not driven by random variation within our models.

In terms of magnitude, the results imply meaningful differences across housing markets. Given the logarithmic factor for both models, the mortgage coefficient of 0.054 corresponds to approximately a 5.4% increase in monthly mortgage payments

attributable to the treatment. This finding supports our hypothesis that the Kalamazoo Promise increased demand for housing, placing upward pressure on home prices, and mortgage costs. In contrast, the treatment coefficient of -0.026 for our rent model implies a 2.6% reduction in rent. This smaller and negative effect suggests that housing costs did not increase uniformly across market segments. The divergence between mortgage and rent outcomes indicates different dynamics between ownership and rental markets.

The control variables behave as expected across both models. Household income has a positive and statistically significant effect on both monthly mortgage payments and monthly rent, indicating that higher income households consume more expensive housing. Housing characteristics, including the number of bedrooms and total rooms, are also positively associated with housing costs. Age exhibits a nonlinear relationship, and household size increases rents but has no statistically significant effect on mortgage payments. Model fit is reasonable, with R^2 values of 0.372 for mortgages and 0.320 for rent.

The positive treatment effect in our mortgage regression is consistent with housing market capitalization. The Kalamazoo Promise increased the attractiveness of the area, raising demand for owner occupied housing and driving up home values. As a result, the benefits of the policy appear to be reflected in higher mortgage payments rather than remaining entirely as direct gains to residents.

The divergence between mortgage and rent effects reflects differences in market structure and adjustment. Owner occupied housing responds more strongly to changes in long term location desirability, while rental markets adjust more gradually. A shift from renting to homeownership may have increased demand in the ownership market while reducing pressure in the rental market, contributing to rising mortgage payments and declining rents.

An additional explanation for this divergence is the difference in time horizons between homeowners and renters. Homeownership represents a long-term investment, where individuals internalize future benefits associated with local policies such as the Kalamazoo Promise. As a result, expected future gains are capitalized into housing prices, leading to higher mortgage payments. In contrast, renters typically face shorter term lease arrangements and are less likely to fully internalize long term policy benefits when making housing decisions. This shorter planning horizon reduces the responsiveness of rental demand to the policy, limiting upward pressure on rents.

Taken together, the results support the hypothesis that the Kalamazoo Promise increased housing prices, as reflected in higher mortgage payments. The negative

effect on rents is less intuitive and suggests that rental markets respond differently to the policy. These findings highlight differences in how policy effects are distributed across housing market segments and reinforce the importance of distinguishing between ownership and rental outcomes.

Conclusion

In this paper, we attempt to answer the question: was there an effect on housing prices in Kalamazoo due to the Kalamazoo Promise, a place-based scholarship introduced in 2005 benefiting students in the Kalamazoo Public School system? This scholarship was an attempt to reverse the city's decline, which had been experiencing urban flight. To address this question, we utilized a DID model, with data drawn from the ACS spanning 2005 to 2024. We examined both mortgage payments and rental payments as response variables, while using other variables within the ACS as explanatory variables.

From our DID model, we estimate that the Kalamazoo Promise increased mortgage payments by approximately 5.4%. To translate this into a dollar value, we make several assumptions. First, we assume a 30-year fixed-rate mortgage at 5%, noting that the average annual mortgage rate over the period was 4.95%. Second, we assume a 20% down payment, as any lower percentage would require private mortgage insurance and increase monthly payments. Finally, we use the average annual mortgage payment in our sample, which is \$1,095. Given these assumptions and our 5.4% estimate, we infer that the increase in housing price is approximately \$13,768.

Our results are consistent with recent papers analyzing the Kalamazoo Promise and similar place-based scholarships. One of the studies we referenced used a DID model and estimated that the treatment effect of the Kalamazoo Promise resulted in a 12.4% to 15.0% increase in housing prices (Beasley & Alam 2025). Another study estimated that residential properties experience a 7% to 12% increase in housing prices within three years of the introduction of a place-based scholarship (LeGower & Walsh 2017). Although our estimated effect is smaller in magnitude, it is directionally consistent with these findings, suggesting that our model performs reasonably well.

Our findings indicate that place-based scholarships positively affect housing prices, though the effect is neither immediate nor fully captures the total benefits of the scholarship. Policymakers considering the implementation of such programs should account for these factors, as they may directly impact constituents through increased property taxes or provide benefits through educational opportunities for their children. Additionally, these policies may indirectly benefit unintended groups. As LeGower and Walsh show, increased housing demand in cities with place-based scholarships tends

to favor higher-priced neighborhoods and higher-quality schools. (LeGower 2017)
Therefore, policymakers should consider targeting areas where such programs can most effectively reduce inequality while supporting long-term economic outcomes.

References

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Appendix

Table 1: Estimated Present Value of the Scholarship for Entering Students

Years in KPS	Tuition Covered	Value of Scholarship
K–12	100%	\$71,915
4–12	90%	\$54,483
6–12	80%	\$44,433
8–12	70%	\$35,671
9–12	65%	\$31,727

Source: Miller (2018)

Table 2. Summary Statistics

Summary Statistics (Mean) by Treatment and Time					
	Control-Pre	Control-Post	Kalamazoo-Pre	Kalamazoo-Post	Total
Mortgage	860	874	844	926	881
Rent	732	897	701	903	882
Household Income	68,629	84,004	67,844	85,798	82,658
Age	42	45	41	43	44

Household Size	3	3	3	3	3
Bedrooms	3	3	3	3	3
Other Rooms	6	7	6	7	7
<i>No. of Observations</i>					139,198

Table 3. Regression Results

	Mortgage	Rent
(Intercept)	4.502*** (0.101)	6.180*** (0.051)
treat	0.054*** (0.011)	-0.026*** (0.007)
lnhincp	0.144*** (0.005)	0.033*** (0.002)
agep	0.004* (0.002)	-0.014*** (0.001)
I(agep^2)	-0.000** (0.000)	0.000*** (0.000)
np	-0.000 (0.003)	0.038*** (0.003)
bdsp	0.097*** (0.007)	0.099*** (0.005)
rmsp	0.040*** (0.003)	0.022*** (0.003)
Num.Obs.	10550	17625
R2	0.372	0.320
R2 Adj.	0.368	0.318
AIC	16074.8	22736.6
BIC	16532.5	23234.4
Log.Lik.	-7974.416	-11304.324
F	101.910	133.347
RMSE	0.46	0.40

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001